国防科技大学
National University of Defense Technology

2025年ICM D题 经验分享

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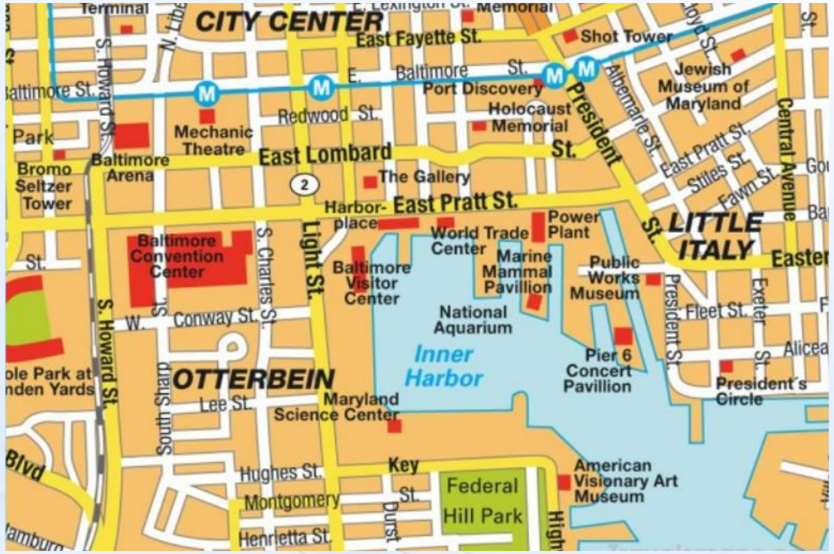
2025年12月12日

汇报提纲

1.1 赛题背景

城市交通是居民日常出行的基本保障。它不仅直接影响通勤效率、生活便利性和出行体验，还对城市经济活力、环境质量和空间布局产生深远影响。高效的交通系统能够减少通勤时间和出行成本，从而提高居民的生活质量。此外，它还是促进城市资源合理配置、推动区域均衡发展以及提升城市吸引力的关键因素。

马里兰州巴尔的摩市面临着严峻的交通挑战，包括基础设施维护和发展资金不足，这使得道路状况恶化，公共交通系统压力增大。该市交通拥堵严重，居民在高峰时段通勤花费大量额外时间。2024 年弗朗西斯·斯科特·基大桥的坍塌等重大事件进一步凸显了其交通网络的脆弱性以及投资的紧迫性。此外，土地利用和交通规划缺乏协调，导致交通分布效率低下。解决这些问题需要采取综合策略，以提高基础设施的抗灾能力，改善公共交通服务，并将可持续交通优先事项纳入城市发展规划。



1.2 数据集

1. **Bus_Routes.csv**:[5] This dataset represents the locations of MTA bus routes within the City of Baltimore as of 2022.
2. **Bus_Stops.csv**:[6] This dataset represents the locations of MTA Bus Stops as of 2022 within the City of Baltimore.
3. **nodes_all.csv**:[7] This dataset represents the locations of tagged geographic attributes by OpenStreetMaps[8] that provide transportation data points in Baltimore. Generally, these are locations where two transportation paths (road, highway, bikeway, walkway, etc.) intersect.
4. **nodes_drive.csv**:[7] This dataset represents the locations of tagged geographic attributes by OpenStreetMaps[8] for car travel. Generally, these are locations where two roads or highways intersect.
5. **edges_all.csv**:[7] This dataset represents the transportation paths between two nodes from the nodes_all.csv dataset.
6. **edges_drive.csv**:[7] This dataset represents the roadways between two nodes from the nodes_drive.csv dataset.
7. **MDOT_SHA_Annual_Average_Daily_Traffic_Baltimore.csv:**[9] MDOT SHA Annual Average Daily Traffic (AADT) data consists of linear & point geometric features which represent the geographic locations & segments of roadway throughout the State of Maryland that include traffic volume information. Traffic volume information is produced from traffic counts used to calculate annual average daily traffic (AADT), annual average weekday traffic (AAWDT), AADT based on vehicle class (current year only) for roadways throughout the State.
8. **Edge_Names_With_Nodes.csv**:[7] This dataset pairs the information from the nodes_all.csv dataset with information from the edges_all.csv dataset to provide street names with nodes.
9. **DataDictionary.csv**: This data file describes the features in each of the data sets provided for this question.

Bus_Routes.csv

巴尔的摩市MTA公交路线的信息

Bus_Stops.csv

巴尔的摩市MTA公交系统的站点详细信息

nodes_all.csv

交通节点位置信息

nodes_drive.csv

机动车交通节点位置信息

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edges_all.csv

连接所有节点的交通路径

edges_drive.csv

连接机动车交通接连的交通路径

MDOT_SHA_Annual_Average_Daily_Traffic_Baltimore.csv

该地区年平均日交通量

Edge_Names_With_Nodes.csv

带有街道名称的节点-边对应关系

1.3 赛题分析

As examples, four transportation issues are outlined in these references:

1. The rebuilding of a collapsed bridge (Francis Scott Key Bridge) in the Harbor.[1]
2. The inadequacy of the minimal public rail systems (MARC, light rail, heavy rail), which connect suburbs that already have several transportation options. The rail transits are not substantial enough to enable commuters and residents to easily use the system to get to the workplace and the downtown free buses primarily help tourists and not the residents of intercity communities.[2]
3. Planning for fixing the disruption over decades on urban communities by US-40 (Highway to Nowhere) through the collaborative West Baltimore United Project.[3]
4. A travelogue of a resident of Brooklyn (community in Baltimore) and his ordeal of trying to use buses to get home after attending a football game in the city.[4]

Using your model, consider projects related to these transportation issues:

1. The collapse of the Francis Scott Key Bridge had a large impact on the transportationsystem of Baltimore. What does your network model(s) show is the impact of the bridgecollapse and/or the reconstruction of the bridge? Be sure to highlight the impacts on thevarious stakeholders in and around Baltimore.
2. Many residents of the City of Baltimore walk or travel by bus. Select a project orpotential project that impacts the bus or pedestrian walkway systems. What does yournetwork model(s) show is the impact of this project? Be sure to highlight the impacts onthe various stakeholders in and around Baltimore.
3. Recommend a project for the transportation network that best improves the lives of theresidents of Baltimore.

   a.What are the benefits to residents of this project?

   b.How does your project impact other stakeholders?

   c.Explain the ways that your project disrupts other transportation needs andpeople's lives.

大桥坍塌对当地的影响

↓

MARC存在不足，出行方式限制性严重

↓

改善该地区交通网络

↓

构建一个网络模型

↓

推荐一个改善项目

汇报提纲

二、求解方法

2.1 实现交通网络流量分配

问题

All of Baltimore's transportation plans affect multiple stakeholders with differing perspectives. Your team's assignment is to improve the lives of the city's residents by recommending ways to improve Baltimore's transportation network.

两个子问题：
- ➢ 交通网络表示
- ➢ 交通流建模

思考

- ➢ 原始数据如何进行处理?
- ➢ 具体交通网络由哪些参数构成?
- ➢ 部分参数指标如何定义?

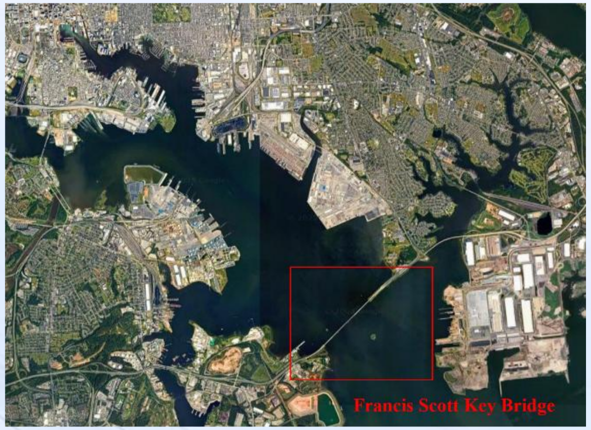
1、利用3-σ原则清洗数据
2、基于题给参数构建交通网络
3、归一化定义参数指标

模型

➢ 子问题一：交通网络表示

➢ 采用多元组、拓扑结构进行建模

$$\begin{cases} G=(V,E,A,W) \\ v_i=\langle ID_i,(lat_i,lon_i),C_i,\{e_j\}_{j\in adj(i)}\rangle \\ \Phi_{ij}=\left[AADT_{ij},AAWDT_{ij},\dfrac{AADT_{ij}}{l_{ij}},\{v_c\}_{c\in V}\right] \end{cases}$$



二、求解方法

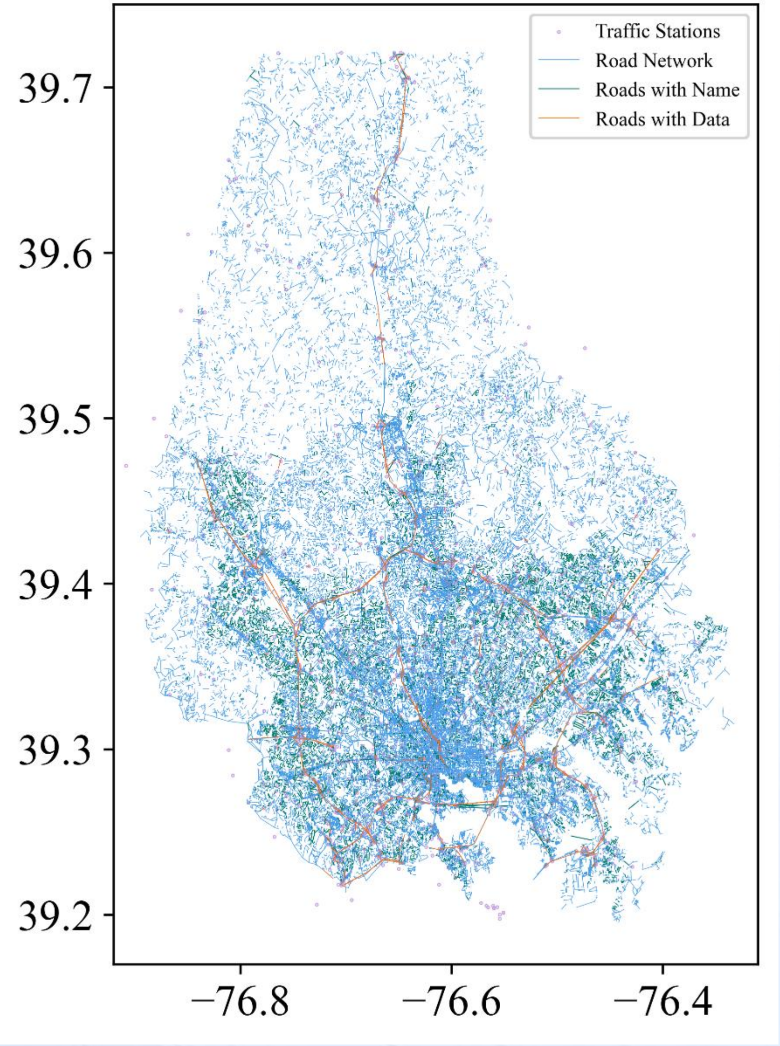
2.1 实现交通网络流量分配

模型

- 子问题二：交通流建模 → 定义表征交通动态基本关系

$$
\begin{cases}
\sum_{e_{ki}\in E_{in}(i)} AADT_{ki} + G_i = \sum_{e_{ij}\in Eout(i)} AADT_{ij} + L_i \\
I(G) = \dfrac{1}{n(n-1)} \sum_{v_i \neq v_j} \dfrac{1}{d(v_i, v_j)} \\
\Delta_G(e_{ij}) = \dfrac{I(G) - I(G/e_{ij})}{I(G)} \\
e_{ij} \leftrightarrow D_k \Leftrightarrow Hausdorff(geom(e_{ij}), geom(D_k)) < \epsilon
\end{cases}
$$

无量纲指标 → 全局度量 → 空间对应关系



2.2 分析弗朗西斯·斯科特·基大桥事故综合影响

问题

1. The collapse of the Francis Scott Key Bridge had a large impact on the transportation system of Baltimore. What does your network model(s) show is the impact of the bridge collapse and/or the reconstruction of the bridge? Be sure to highlight the impacts on the various stakeholders in and around Baltimore.

两个子问题：
- 对交通流量的影响
- 对居民的影响

思考

- 如何评判对交通流量的影响?
- 居民出行成本如何定义?
- 如何进

1、基于AADT方差选取主干道进行评判
2、利用Wordrop均衡原理确定出行时间
3、基于模糊综合评价架构四维因子空间

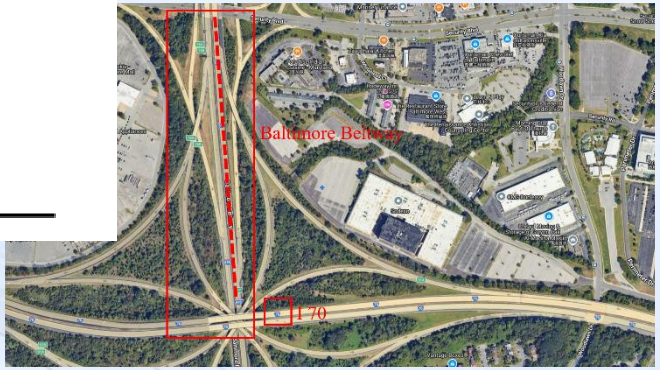
弗朗西斯·斯科特·基大桥坍塌后交通方差对主要道路的影响

	Station ID	AADT Variance		Station ID	AADT Variance
1	B1110	25319.40571	5	B1111	9904.712673
2	B1105	15883.16452	6	B1174	8138.236222
3	B240002	12513.39594	7	B0983	7922.138179
4	B240004	9931.978884	8	P0071	7922.138179

模型

- 子问题一

$c_{e_b}(t) = \left\{ \right.$ … $t_{collapse}$



二、求解方法

2.2 分析弗朗西斯·斯科特·基大桥事故综合影响

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两个子问题：
- 对交通流量的影响
- 对居民的影响

思考

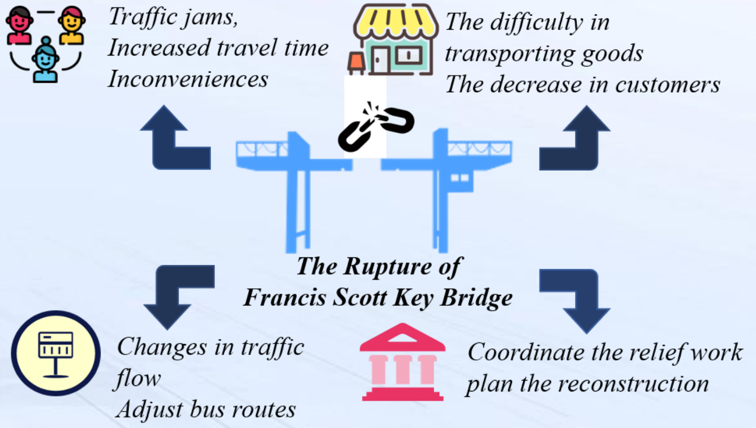
- 如何评判对交通流量的影响?
- 居民出行成本如何定义?
- 如何进行模型建立?

1、基于AADT方差选取主干道进行评判
2、利用Wordrop均衡原理确定出行时间
3、基于模糊综合评价架构四维因子空间

模型

- 子问题二：对居民的影响
- 基于Wordrop均衡原理衡量

1.在均衡状态下，所有被使用的路径具有相同且最小的旅行时间
2.在均衡状态下，交通流在网络中的分配使得整个系统的总旅行时间最小化。



二、求解方法

2.3 对多类出行方式进行合理安排

问题

2. Many residents of the City of Baltimore walk or travel by bus. Select a project or potential project that impacts the bus or pedestrian walkway systems. What does your network model(s) show is the impact of this project? Be sure to highlight the impacts on the various stakeholders in and around Baltimore.

两个子问题：
- 多模式网络表示
- 需求分配

思考

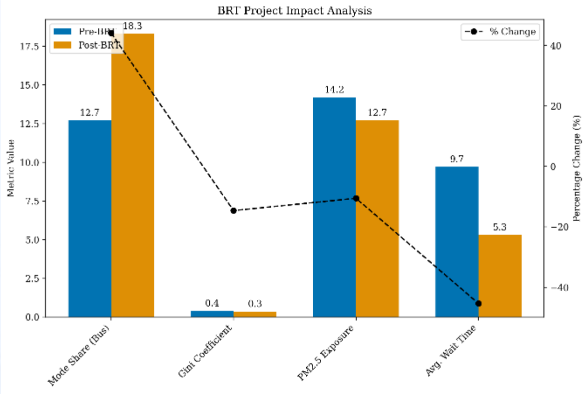
- 多式联运系统如何进行数学上的形式化？
- 出行需求包含哪些相关效用？
- 如何体现多模式系统的改善？

1、建立多层网络模型
2、由嵌套逻辑模型决定需求分配
3、结合具体场景快速交通(BRT)项目实施

模型

- 子问题一：多模式网络表示
- 形式化为三层有向图

$$
\begin{cases}
G_M=(N,\varepsilon,M) \\
N=N_b\cup N_p\cup N_v;\varepsilon=E_b\cup E_p\cup E_{trans};M=\{\text{公交、步行、汽车}\} \\
t_b(e_{ij}^b)=t_\omega(f_j)+motion\ time+congestion\ penalty \\
c_p(e_{kl}^p)=\dfrac{l_{kl}^p}{1.34}+\sum_{m=1}^{M}\omega_m d_m(kl)+\lambda\left[1+e^{-\left(\frac{V}{C_{ped}}-0.8\right)}\right]^{-1}
\end{cases}
$$



2.3 对多类出行方式进行合理安排

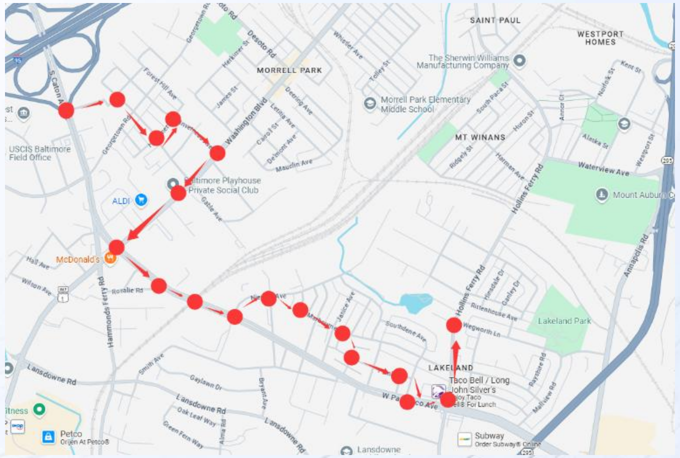
模型

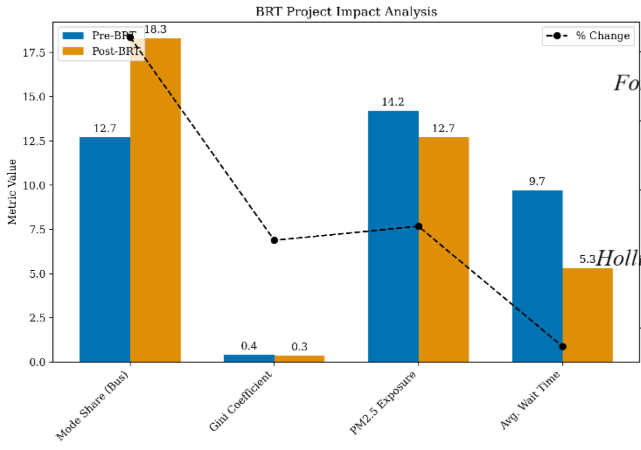
- 子问题二：需求分配 → 基于嵌套逻辑模型决定

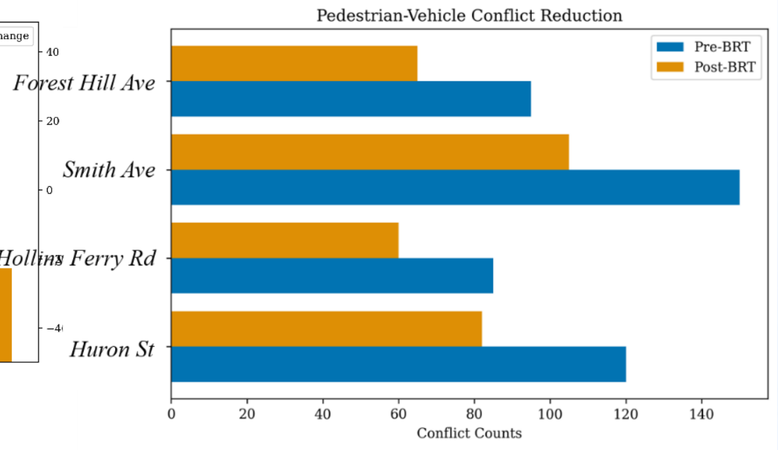
$$
\begin{cases}
P_{m|od} = \dfrac{\exp\left(V_{od}^{m} + \ln S_m\right)}{\sum\limits_{m'} \exp\left(V_{od}^{m'} + \ln S_{m'}\right)} \\
V_{od}^{m'} = \beta_{cost} \cdot c_{od}^{m} + \beta_{time} \cdot t_{od}^{m} + \beta_{rel} \cdot \sigma_{od}^{m} \\
S_m = \sum\limits_{r \in R_{od}^{m}} \exp\left(\mu_m \left(V_{od}^{r} - V_{od}^{m}\right)\right)
\end{cases}
$$

Metric	Pre-BRT	Post-BRT	Δ%
Mode share (bus)	12.7%	18.3%	+44.1%
Accessibility Gini	0.41	0.35	-14.6%
PM2.5 exposure (μg/m³)	14.2	12.7	-10.6%
Avg. wait time (min)	9.7	5.3	-45.4%

实施场景







二、求解方法

2.4 案例研究

问题

3. Recommend a project for the transportation network that best improves the lives of the residents of Baltimore.
   a. What are the benefits to residents of this project?
   b. How does your project impact other stakeholders?
   c. Explain the ways that your project disrupts other transportation needs and people's lives.

项目建议：综合移动走廊

三目标优化范式

$$\max_{x\in\chi}\left[f_1(x), -f_2(x), f_3(x)\right]$$

$f_1(x) \longrightarrow$ 居民福利提升

$-f_2(x) \longrightarrow$ 实施成本最小

$f_3(x) \longrightarrow$ 环境效益最大

效益量化模型

嵌套逻辑模型 $\longrightarrow P_{m|od}$

量化广义出行成本降低

量化人口健康改善

逻辑扩散

经济可达性改善

系统中断分析

临时网络退化

嵌套逻辑模型

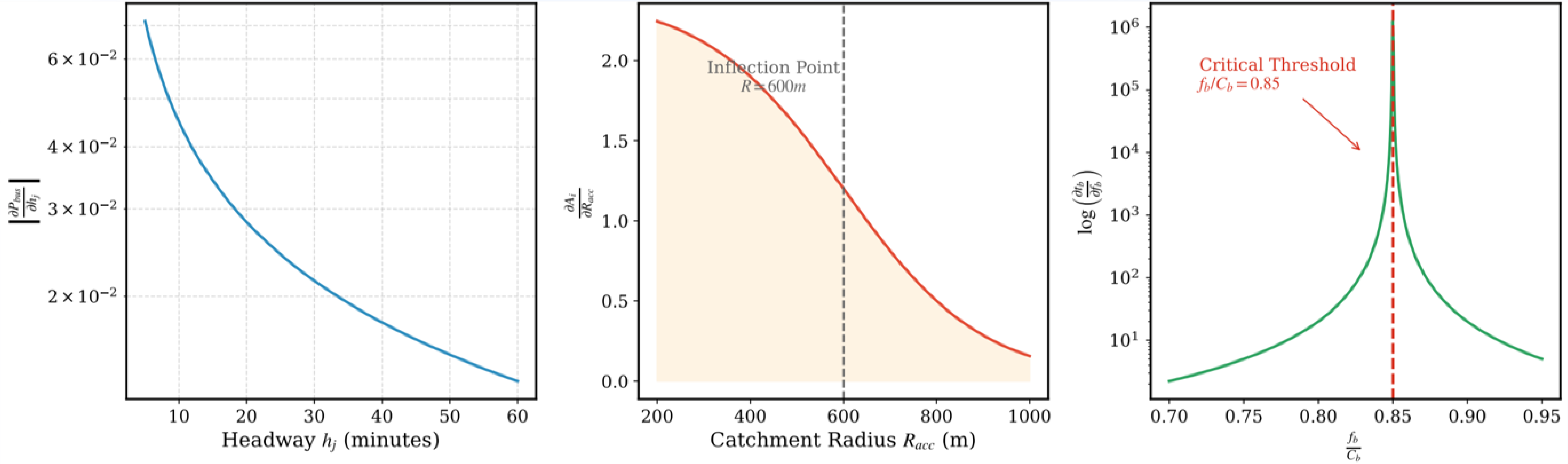
$$\begin{cases} c_e(t) = c_e^0 \cdot \left[1 + \kappa \cdot sigmoid(t_{start}, t_{end})\right] \\ \Delta c_{auto} = \dfrac{x_b}{L_{tot}} \cdot \left(\dfrac{q_{auto}}{q_{auto} - \Delta q_{mode}}\right)^{\beta} \\ G(t) = G_0 e^{-\lambda t} + \dfrac{\theta \Delta A}{1 + e^{-(t-t_0)/\tau}} \end{cases}$$

案例研究:巴尔的摩绿色交通走廊

二、求解方法

2.5 灵敏性分析

1. 公交分担率对发车间隔敏感：公交分担率随发车间隔的缩短呈幂律增长，说明缩短间隔可提升公交使用率。
2. 可及性收益与接驳半径呈S型衰减：其在接驳半径约600米处收益增长最快。
3. 系统存在不稳定阈值：当公交负载率f_b/C_b >0. 85时，边际阻抗急剧增大，系统接近崩溃点。



(a) Bus Mode Share Sensitivity (b) Accessibility Sensitivity (c) Congestion Collapse Threshold

汇报提纲

3.1 摘要

结构设计：三步切割法+五要素法（问题+方法+创新点+结果+结论）

首段定基调（2-3句）

☑ 合格版：

"This paper addresses the multi-objective optimization problem of wildfire evacuation routes under dynamic wind conditions. By integrating real-time meteorological data and crowd mobility patterns..."

✕ 低分版：

"This paper studies wildfires. We consider wind and people. Our model is useful for solving this problem."

3.1 摘要

方法论快照（4-5句）

"A dynamic decision framework (Eq.1) is proposed, where the risk index R is calculated by combining flame spread rate and population density."

公式密度控制：摘要中最多出现“一半加一”个核心公式

3.1 摘要

结构设计：三步切割法+五要素法（问题+方法+创新点+结果+结论）

结论记忆点（2-3句）

用数据强化可信度：

"Simulation shows response time reduced by 37.5% compared to static models, validated by historical datasets from 2018-2022 California wildfires (MAPE=6.8%)."

汇报提纲

4.1 赛前准备

01

组队情况

专业能力互补

电子对抗+仿真+大数据

角色经验互补

编程手 → 建模手 → 写作手

02

备战情况

具备数模经验

24美赛B题M、24国际物理竞赛银奖、军事建模一等奖

参加数模培训营

阅读获奖论文、学习经验、积累各类工具

03

比赛计划

确定场地和时间

最好线下，便于交流，并且每天固定工作时间

确保进度合理

提前对解题有大致规划

建议J人P人搭配

4.2 赛中过程

问题数学化
实际问题转为数学问题。

建模型定方法
确立模型结构，选择求解路径。

计算验证
求解模型结果，验证初步有效。

调优反馈
分析结果调整模型，重新验证。

论文撰写
撰写模型报告，清晰呈现结论

四、参赛经历

4.3 赛后总结

1

2

3

4

团结

- 合理分配工作
- 严格规划时间
- 树立共同目标

耐心

- 合理分配精力
- 坚持到最后
- 保证充足休息

创新

- 学习往年论文
- 突出工作特色
- 给评审留印象

严谨

- 模型的合理性
- 结果可解释性
- 研究的现实意义

感谢聆听

祝大家取得满意的成绩！

衷心感谢各位同学，请批评指正！